INFERTILITY

Definition:

Inability to conceive after 1 year of continuous marital state with regular intercourse &without use of contraception.

* ***Inability to conceive:*** we have 2 items.

1- ***Sterility:*** inability to conceive

2- ***Infertility:*** inability to get viable offspring including cases of habitual abortion. Others say that infertility: inability to conceive due to curable (relative) cause e.g. tubal adhesion, while sterility due to incurable cause e.g. absent uterus

* ***After 1 year:*** why 1 year? → fecundability: is the monthly probability of couple to get conception.

After 1 cycle of marriage → 20-25%

After 1 year of marriage → 95 – 100 %.

* ***So in definition:*** of 1ry infertility → after 1 year while in secondary infertility → we add other 1 year of lactational anovulation (physiological) so secondary infertility → after 2 years.

Primary infertility: Never conception.

Secondary infertility: Previous conception even with abortion, ectopic, vesicular mole.

* ***Regular sexual life:*** In both of:

a- Frequency 3-4 / week at time of ovulation.

b- *Technique:* erection → penetration → ejaculation of semen in posterior fornix (semen pool).

Incidence: 10-15% of all women, with ↑ incidence nowadays why:

1- ↑↑ incidence of sexually transmitted diseases.

2- ↑↑ age of marriage (note that peak of female fertility at 24 year)

3- air pollution.

4- ↑↑ methods of diagnosis.

Incidence:
- → Male factor 30%
- → Female factor 50%
- → Disturbed sexual life 10%
- → Unexplained factor 10%

Aetiology : **I- <u>Male factor</u>: 30%**

Defect in Male: 1- Defect in semen production.

2- Bilateral vas obstruction.

3- Failure of semen deposition in seminal pool.

A- <u>↓in semen production: abnormal semen production</u>

These due to:

1- Congenital cause
- → Absent testis
- → Undescended testis

2- Inflammatory → Mumps, measles

3- Traumatic
- → Accidental injury of testis
- → Surgical removal

4- Tumours of testis: teratoma

5- Irradiation

6- Varicocele: elongated, dilated, tortous pampiniform plexus of veins around testis, lead to

↓ Hyperthermia

↓ Hypoxia → ↓ of the testes

↓ Toxins

7- Physical factors: Of little value. e.g. hot baths, tight underwear

B- <u>*Bilateral vas obstruction*</u>:

1- **Congenital:** Aplasia or hypoplasia of both vas.

2- **Inflammatory*:*** Gonorrhea and chlamydia

3- **Traumatic:** Bilateral vas ligation in repair of bilateral inguinal hernia.

4- **Tumours:**

5- **Irradiation**

C - Failure of deposit of semen in posterior fornix:

1- **Impotence:** Failure of male erection.
2- **Premature ejaculation:**
3- **Retrograde ejaculation:** Most of ejaculate pass to bladder and not through urethra to outside due to defect in valve system between urethra and bladder.
4- **Hypospaduis:** Urethral opening on ventral aspect of penis
5- **Epispadius:** Urethral opening on dorsal aspect of penis.

II- Female Factor: 50%

***A- Ovarian cause:* 15-20%**

1- **Causes of an-ovulation*:*** = causes of hypothalamus, pituitary and ovarian amenorrhea (mention).
2- **Causes of luteal phase defect (LPD)**

***B- Tubal factor*: 20%**

1- **Congenital:** Hypoplasia or aplasia.
2- **Inflammatory:** Salpingitis*:* most common is chlamydia & gonorrhea.
3- **Traumatic:** Surgical removal
4- **Endometriosis.**
5- **Tumors:**

***C- Uterine factor*: 5%**

1- **Congenital:** hypoplasia; or aplasia.
2- **Inflammatory*:*** e.g. T.B. endometritis.
3- **Traumatic:** Asherman's syndrome.
4- **Neoplasms*:*** Either benign → fibroid
   malignant → endometrial cancer.

***D- Cervical factor:* 5%**

1- Congenital*:* → Cervical stenosis
→ Cervical elongation

2- Inflammatory*:* Chronic cervicitis.

3- Cervical displacement: e.g. RVF, prolapse.

4- Immunological infertility : Presence of cervical sperm antibodies in cervical mucus.

5- Neoplasms: Either benign → cervical fibroid or malignant → cancer cervix.

***E- <u>Vaginal factor</u>*: 1%**

1- Congenital*:* Aplasia or hypoplasia or vaginal septum.

2- Inflammatory: Vaginitis.

3- Traumatic: Surgical removal.

4- Neoplasms*:* Either benign or malignant.

F- <u>General factor</u>: Act through inhibition of ovulation

1- Endocrine disorders: e.g. D.M., myxoedema.

2- T.B.

3- Anemia. 4- Psychological disorders.

III- <u>Disturbed sexual life</u>: 10%

1- Too frequent or too infrequent coitus.

2- Dyspareunia: difficult or pain during or after coitus.

3- Vaginismus: violent reflex spasm of levator ani, adductor of thigh, gluteal muscles and anterior abdominal wall muscles on any attempt at sexual intercourse.

4- Frigidity: lack of sexual desire.

5-Effluvium seminis: excessive escape of seminal fluid from vagina after coitus.

IV- <u>Unexplained infertility</u>: 5-10%.

Infertility with no apparent cause with examination and routine investigation for both partners.

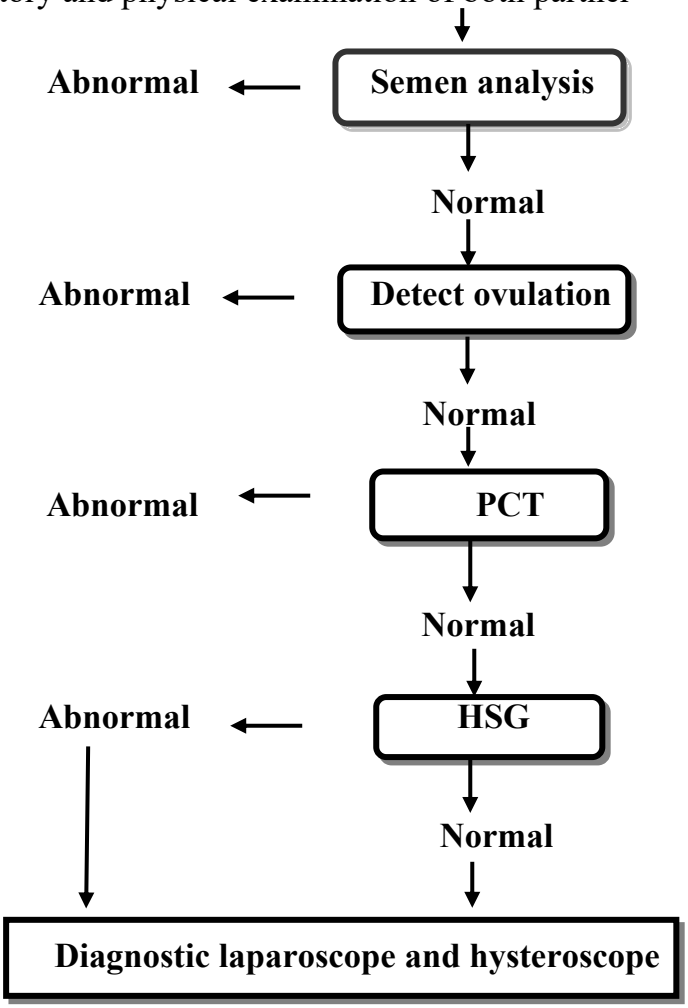
***<u>Causes</u>*:** 1- Atypical endometrioss. 2- Ovum dysfunction .

3- Sperm dysfunction. 4- Subclinical infection.

5- Subclinical pregnancy loss. 6- Immunological factor.

Investigations of Infertility:

- Complete history and physical examination of both partner



[I] <u>Complete history and physical examination of both partners</u>:

A- *<u>History and examination of female:</u>*

1-History: I- *Personal history:* e.g. age, length of marriage duration and types of infertility and if husband had children from previous marriage.

II-*Past history:* e.g. previous pelvic operation, previous history of PID.

III- *Family history:* of D.M. or T.B. or hypertension.

IV- *Menstrual history:* any abnormalities of cycle,

V-*Obstetric history:* especially puerperal sepsis.

VI- *Sexual history:* - frequency of coitus.

- History of dyspareunia, vaginismus, effluvium seminis.

2-Examination: I- ***General examination:*** e.g. body built, development of secondary sexual characters, hair distribution, measure ABP.

II- ***Abdominal and local examination:*** may detect pathology which cause infertility e.g. fibroid, PID., hypoplastic uterus.

B) ***History and examination of male*****:**

1- History: * Previous marriage and getting of children.

* History of mumps, gonorrhea.

* Previous pelvic operation e.g. hernia, varicocele.

2- Examination: I- ***General:*** for any endocrine diseases.

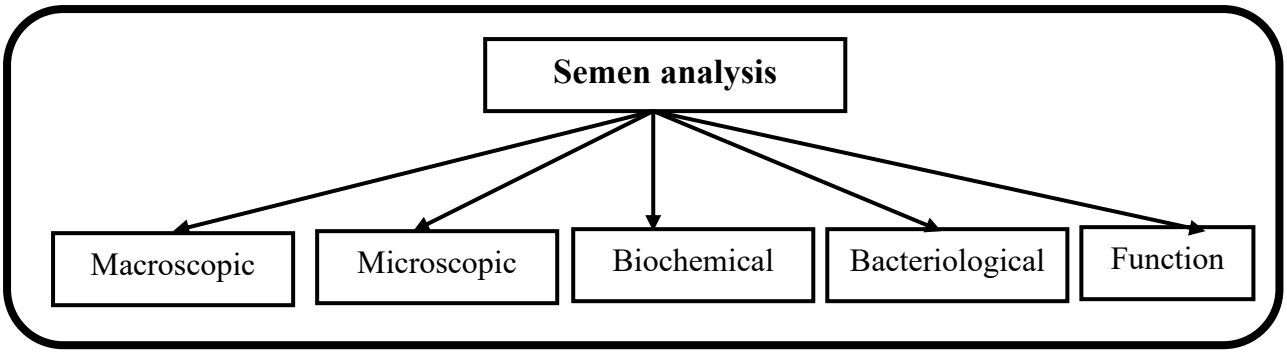
II- ***Abdominal and local examination:*** Scar of previous operation, hypospadius, urethral discharge, nodules of epididymis (T.B.) or varicocele (bag of warms).

[II]- Investigations of male factor:

(A) Semen Analysis:

- By 2 systems: seminogram or computerized assisted of semen analysis.
- After 2 – 3 days of no coitus.
- Collect specimen by one of 2 methods masturbation or coitus interuptus.
- Repeated 2 separate times, 2 months apart.
- Examination in 2 hr (Motility : at once, after 1 hour, 2 hour).

WHO criteria for normal seminogram:	Abnormal
Ejaculate must be 2 – 6 ml.	Aspermia: no semen Hypospermia: < 2 ml Hyperspermia : > 6 ml
Count 20 – 60 million / ml.	Azoospermia: no sperm Oligospermia: < 20 million Polyspermia : > 200 million
More than 2 sperms move progressively forwards.	
Motility > 60% at 2hr after ejaculate.	Asthenospermia: weak Necrospermia: dead
Normal forms > 60% .	Teratospermia: ↑ abnormal forms
No significant pyospermia (+ve pus cells).	
No significant hyperviscosity.	
No significant sperm agglutination.	



[I] Macroscopic Criteria:

1- Liquifaction: Normal in 20 min.

If delayed $> \frac{1}{2}$ hr : ↑ viscosity. $< \frac{1}{2}$ hr : ↓ viscosity.

2- Consistency: Viscid which liquefies in 20 – 30 min.

3- Colour: Whitish or opalescent (Milk colour).

4- Odour: Acridic.

5- Reaction: Alkaline : 7.4 pH.

6- Volume: (2-4 ml)

No : aspermia.

< 2 ml : hypospermia.

> 4 ml : hyperspermia.

[II] Microscopic criteria:

1- Sperm count: N = 60 – 100 million / ml

No sperm : azospermia.

< 20 million / ml : oligospermia.

< 5 million / ml : severe oligospermia.

> 200 million / ml: polyspermia.

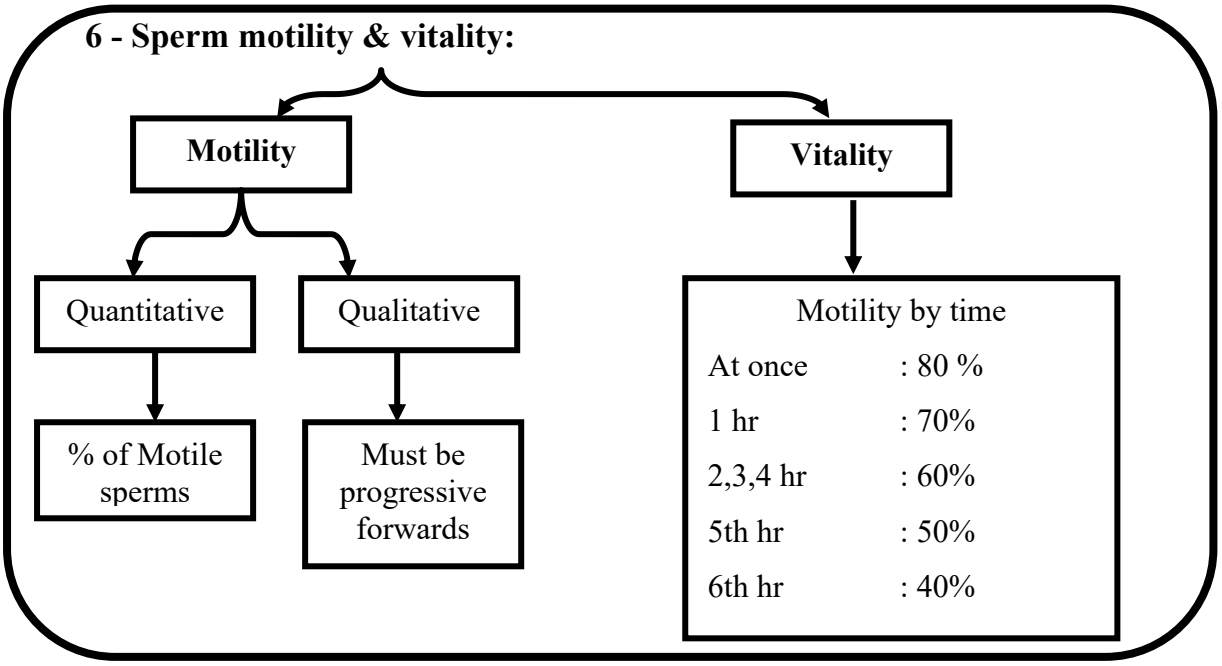
2- Sperm morphology:

Normal forms must be >60%. , Abnormal forms 20 – 40%.

3- Sperm agglutination: If > 10–20% means, Infection or immunological problems.

4- Pus cells: N. < 5 / HPF. , If > 5 / HPF : Pyospermia.

5- RBC : N < 5 / HPF. , If > 5 / HPF : haenospermia.



Athenospermia: weak sperms : due to infection or immunologic.

Necrospermia : All sperm dead: due to infection.

[III] Biochemical exam:

1- Sperm conc. & ↑ Hepatocyte growth factor / scatter factor ↑ (HGF/SF).

2- Sertoli cell function & ↑ Transferrin.

3- Epididymis & ↑ α - glucosidase.

4- Seminal vesicle & ↑ fructose.

5- Prostate & ↑ acid phosphatase.

6- Infection : ↑ PG & IL-6.

These markers can differentiate site of obstruction in case of obstructive azospermia:

1- Azoo + normal level of fructose & acid phosphatase : obstruction above seminal vesicles.

2- Azzo + ↓ fructose with normal acid phosphatase obstruction ( ) seminal vesicles & Prostate.

3- Azzo + no fructose or acid phosphatase : Obstruction below prostate.

[IV] Bacteriological exam: If infection is suspected do C. & S.

[V] Test for sperm function:

1- Post coital test (PCT).

2- Sperm, Cx mucus contact test (SCMC – test).

3- Cross over test.

4- Hypo-osmotic swelling test.

5- Acrosome reaction (A.R.).

6- Hamster egg. penetration test.

***B) Testicular biopsy*:** to differentiate azoospermia due to testicular damage or bilateral vas obstruction.

If biopsy contains +ve sperms: no testicular damage.

C)Vasography: injection of dye into vas and X-ray to denote site of obstruction.

D) Urine analysis after ejaculation: to determine retrograde ejaculation.

E) Hormonal assay: e.g. testosterone, LH, FSH, in serum.

[III]- Detection of ovulation:

***History*:** - Regular cycles.

- Mid ovulatory pain (Mittleschmerz pain).
- Mid ovulatory bleeding.
- Mid ovulatory cascade
- Premenstrual mastalgia.
- Premenstrual tension and headache.
- Spasmodic dysmenorrhea.

***Basal body temperature chart(BBT-chart)*:**

Technique:

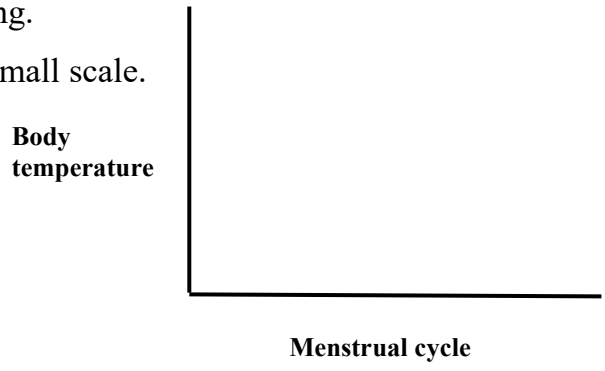
a. Taking temperature early in morning.

b. Special thermometer with special small scale.

c. At least 3-5 min. in mouth.

d- Begins with 1st day of cycle.

e- Every day and recorded on chart.



Value:

1- Ovulation (biphasic chart) which detected by sharp rise of temperature by 0.5°C at midcycle and continues for 14 days due to thermogenic effect of progesterone

2- Detect of anovulation (Monophasic curve)

3- Detect of LPD (Short second phase).

4- Help in timing the intercourse.

Premenstrual endometrial biopsy: **(PEB)**

Time*:* on day 22 of cycle

Some authors → 3 days before next cycle, others considers first day of cycle

Value*:* 1- Detect ovulation.

2- Dating of endometruim and detect LPD.

3- Detect T.B. endometritis.

***Study of Cx. mucus*:**

*-ve thread test.
*-ve spinnbarkeit test.
*-ve fern test.
} In second half of cycle (22th)

***Study of vaginal cytology*:** ↑ intermediate cells & ↑ leucocytes with ovulation which can be detected in second half of cycle (22th)

U/S: Serial folliculometry around time of ovulation on day 11,13,15, of cycle & ovulation can be detected by:1- Disappearance of mature follicle.

2- Free fluid in Douglas pouch.

3- Presence of corpus luteum.

Hormonal assay:

- Serum L.H. peak at midcycle↑ 100 mU/mL.
- Serum progesterone on day 22 of cycle ≥ 10ng /mL. If serum progesterone 2-10 ng/mL → LPD.

Endoscopy: To visualize process of ovulation which manifested by stigmata on surface of ovary (small trauma)

N.B.: Investigations of LPD:

1- Dated endometrial biopsy.
2- B.B.T. chart.
3- Serum progesterone 2- 10ng/mL.

[IV] Detection of Tubal Patency:

a- Insufflation (Rubin's test)
b- Kymography
c- Hysterosalpingography
d- Chromotubation
e- Hysterosonography :
f- Falloposcopy.

A) *Insufflation* (Rubin's test):

- **Principle:** Injection of CO_2 under pressure into uterus by special cannula and if tubes are patent, CO_2 will pass through then into peritoneal cavity.

N.B.: We not use air to avoid possibility of air embolism caused by nitrogen content, but CO_2 is easily soluble and rapidly excreted by lung.

- **Timing of method:** Best time is post menstrual e.g. (4-7) days after end of menstrual blood. Why?
  a- In this time the endometrium is thin and less vascular, so ↓ incidence of air embolism.
  b- To avoid false –ve results (if done premenstrual we can show bilateral tubal block due to thick endometrium and not due to real blockade).
  c- To avoid endometriosis.
  d- To avoid disturbed pregnancy.
- **Criteria of tubal patency:**
  a. Drop in pressure in manometer.
  b. Auscultation of bubbling sounds over lower abdomen (Hissing sounds).
  c. Shoulder pain: due to irritation of phrenic nerve.
  d. X-ray can show subdiaphragmatic pneumoperitoneum (air under diaphragm).
- **Contraindications:**
  a. pelvic infection.

b. During menstruation.
c. During period of Amenorrhea (as patient may be pregnant)
d. In premenstrual period.

- Side effects:

a. Pelvic infection.
b. Air embolism.
c. Disturbed pregnancy.
d. Sometimes vomiting and collapse due to irritation of peritoneum.

B) *Kymography*: (Sharman's kymography):

In Rubin's test, we determine if tubes patent or not while in kymography we determine if tubes patent or not and study pressure changes and peristalsis waves of tubes by record the osscilations of pressure manometer on rotating drum. So we can differentiate between tubal occlusion or tubal spasm or tubal stenosis.

C) *Hysterosalpingogram*:

Hysterosalpingogram → film.

Hysterosalpingography → technique

- **Principle:** injection of dye through cervix which outline cavity of uterus, both tubes and peritubal area and ,take X-ray film.
- **Timing:** the same as Rubin's test (postmenstrual).
- **Material:** 40-60% iodine.

In oily medium (Lipidol)	In water soluble medium (Urograffin)
1- More detailed information on uterine cavity, tubes and peritoneum	1- less detailed information
2- More therapeutic effect	2- less therapeutic effect
3- ↑↑ incidence of oil granuloma	3- ↓↓ incidence of granuloma
4- ↑↑ incidence of oil embolism	4- ↓↓ incidence of embolism
5- delayed film after 24hr.	5- delayed film after 15 min.30min

- Technique*:*

1- Antispasmodic to ↓ tubal spasm.

2- No anaesthesia.

3- X-ray films:

a- Control film: before inject dye to show any abnormalities of pelvic bones or soft tissue shadow.

b- HSG: X-ray after inject 8-10CC lipidol to outline uterus and tubes.

c- Delayed film 24th later to show peritoneal spill due to scattering of dye by intestinal movements, if tubes are patent.

-Advantage of HSG: over Rubin's test:

1) *Revealed any uterine abnormalities.*

a. Congenital anomalies e.g. bicornuate uterus.
b. Submucous fibroid: → filling defect.
c. Intrauterine adhesions → scattered angled filling defect.

2) *Localizaion site of tubal obstruction:*

a. If unilateral or bilateral.
b. If cornual or midsegment or terminal (fimbrial).

3) *Delayed film: shows:*

a. No peritoneal spill if bilateral tubal occlusion.
b. Accumulation of dye in peritubal area suggesting peritubal adhesion.

4) *Therapeutic effect: due to:*

a- Hydrostatic pressure (fluid pressure) can overcome site of obstruction inside tube.
b- Tubal lavage.
c- Overcome thin peritubal adhesions by iodine.
d- Iodine enhances function of tubal cilia.
e- Iodine enhances characteristics of cervical mucus.
f- Iodine is bacteriostatic.

-Contraindications: as contraindication for Rubin's test+ allergy to iodine.

-Side effects: as side effects for Rubin's test + allergic reaction to iodine.

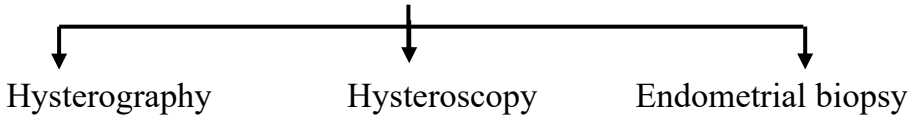
D) *Chromotubation*: *(M.B. test):*

We inject methylene blue (M.B.) dye and visualization of dye coming from both tubes through laparoscopy.

E) *Hysterosonography*: Injection of saline through cervix & visualize it coming from fimbria by sonar.

F) *Falloposcopy*: Endoscopic visualization of interior of tubes

[V] Investigation of uterine factor:



[VI] Investigations of cervical factor:

A) *Study of cervical Mucus in pre-ovulatory period*: to be suitable for sperm penetration, cervical mucus should be profuse, thin, clear, acellular, alkaline, maximum ferning and maximum thread test (length of thread about 10cm).

B) *Postcoital test*: - Should be preceded by semen analysis

1- Fractional postcoital test.

- Best done at time of ovulation to study cx. mucus with it.
- A drop of discharge is taken from posterior fornix and another drop from Cx. mucus within 6hr after coitus and both examined microscopically:

Vaginal drop	Cx. drop	Result
No spermatozoa	No spermatozoa	Failure of deposition
All sperms dead	No spermatozoa	↑↑ vaginal acidity
Living sperms	Dead spermatozoa	Hostile cx. mucus (infection)
Living sperms	Living spermatozoa	Normal

2- Sim's Huhner PCT:

- Cx mucus exam by HPF of microscope & assess sperm motility & count score:

0 = 0 / HPF (No sperm found).

Bad	1	= any count but not motile.
	2	= 1 – 5 motile sperms / HPF
Normal	3	= 6 – 10 motile sperms / HPF
	4	= 11 – 20 motile sperms / HPF
	5	= ≥ 20 motile sperms / HPF

The motility must be progressive forwards as shaking movements may due to infection or antisperm A.B.

3-Laparoscopic PCT (Sperm recovery test):

Laparoscopic retrieval of sperms from D.P. after coitus.

C) *Sperm cervical mucus test (SCMC)*:

Done when PCT is –ve

1- Slide SCMC – Test: (Miller – test)

Mid cyclic CX mucus drop & sperm drop on a slide both covered with cover slip & exam the interface ( ) sperms & Cx mucus.

Slide show if +ve penetration of Cx mucus or not.

2- Capillary tube test: (Kremer – test)

Sperms placed in reservoir & mucous sucked in Capillary tube fixed on scaled slide at 37°C & exam. after, 30 min, 3, 24 hr for mucous penetration.

[V] Investigation of vaginal factor:

- Vaginal cytology to detect type of vaginitis

[VI]- Investigation of general factor:

a- Hormonal assay: TSH, ACTH, prolactin, androgen.

b- Hb % c- Blood sugar.